

Yuxuan Ren

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EDUCATION

Washington University in St. Louis <i>Ph.D. Candidate, Environmental Engineering</i>	St. Louis, MO 08/2022 – Present
Peking University <i>M.S., Environmental Science</i>	Beijing, China 09/2019 – 06/2022
Shandong University <i>B.E., Environmental Engineering</i>	Jinan, China 09/2015 – 06/2019

RESEARCH INTERESTS

I am an atmospheric chemist. I use ground-based measurements and chemical transport models to study atmospheric composition, inform air quality policy, and protect human health.

PROFESSIONAL EXPERIENCE

Research Assistant , Washington University in St. Louis Advisor: Randall V. Martin	01/2023 – Present
• Characterized aerosol composition using the globally distributed Surface Particulate Matter Network (SPARTAN) • Evaluated global black carbon emissions with SPARTAN measurements and chemical transport model GEOS-Chem • Developed an offline aerosol mass spectrometry method for organic aerosol characterization on PTFE filters • Developing a global organic aerosol dataset for model evaluation	
Research Assistant , Peking University Advisor: Hefa Cheng	06/2018 – 06/2022
• Characterized heavy metals in particulate matter (PM) across 12 major cities in China • Assessed the bioaccessibility and health risks of PM-bound heavy metals using simulated physiological fluids	

TECHNICAL SKILLS

Atmospheric measurement: mass spectrometry (HR-ToF-AMS, ICP-MS, GC-MS, HPLC-MS) and other aerosol instruments (SMPS, XRF, IC, TOC, UV-Vis)

Atmospheric modeling: global atmospheric chemistry (GEOS-Chem)

PEER-REVIEWED PUBLICATIONS

H-index: 7, citations: 217 (as of 07/2026, Google Scholar)

First-Author Papers

[4] Ren, Y., Martin, R. V., Chen, J.-H., Oxford, C. R., Williams, B. J., Weber, R. J., Xu, L. Development of an Offline Aerosol Mass Spectrometry Method for Organic Aerosol Characterization in a Globally Distributed Surface Particulate Matter Network. *EGUsphere* [preprint], 2026.

<https://doi.org/10.5194/egusphere-2026-2663>

[3] Ren, Y., Oxford, C. R., Zhang, D., Liu, X., Zhu, H., Dillner, A. M., White, W. H., Chakrabarty, R. K., Hasheminassab, S., Diner, D., Le Roy, E., Kumar, J., Viteri, V., Song, K., Akoshile, C., Amador-Muñoz, O., Asfaw, A., Chang, R. Y.-W., Francis, D., Gahungu, P., Garland, R. M., Grutter, M., Kim, J., Langerman, K., Lee, P.-C., Lestari, P., Mayol-Bracero, O. L., Naidoo, M., Nelli, N., O'Neill, N., Park, S. S., Salam, A., Sarangi, B., Schechner, Y., Schofield, R., Tripathi, S. N., Windwer, E., Wu, M.-T., Zhang, Q., Rudich, Y., Brauer, M., Martin, R. V. Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. *Nature Communications*, 16, 7010, 2025.

<https://doi.org/10.1038/s41467-025-62468-5>

[2] Ren, Y., Hu, Y., Cheng, H. Sources, Bioaccessibility and Health Risk of Heavy Metal(loid)s in the Particulate Matter of Urban Areas. *Science of the Total Environment*, 947, 174303, 2024.

<https://doi.org/10.1016/j.scitotenv.2024.174303>

[1] Ren, Y., Luo, Q., Zhuo, S., Hu, Y., Shen, G., Cheng, H., Tao, S. Bioaccessibility and Public Health Risk of Heavy Metal(loid)s in the Airborne Particulate Matter of Four Cities in Northern China. *Chemosphere*, 277, 130312, 2021.

<https://doi.org/10.1016/j.chemosphere.2021.130312>

Co-Authored Papers

- [6] Liu, X., Turner, J. R., Mitroo, D., **Ren, Y.**, Oxford, C. R., Liu, W., Martin, R. V. Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust. *ACS ES&T Air*, **3**, 175–185, 2025.
<https://doi.org/10.1021/acsestair.5c00295>
- [5] Luo, Q., **Ren, Y.**, Sun, Z., Li, Y., Li, B., Yang, S., Zhang, W., Wania, F., Hu, Y., Cheng, H. Characterization of Atmospheric Mercury from Mercury-Added Product Manufacturing Using Passive Air Samplers. *Environmental Pollution*, 337, 122519, 2023.
<https://doi.org/10.1016/j.envpol.2023.122519>
- [4] Wang, X., **Ren, Y.**, Yu, Z., Shen, G., Cheng, H., Tao, S. Effects of Environmental Factors on the Distribution of Microbial Communities Across Soils and Lake Sediments in the Hoh Xil Nature Reserve of the Qinghai-Tibetan Plateau. *Science of the Total Environment*, 838, 156148, 2022.
<https://doi.org/10.1016/j.scitotenv.2022.156148>
- [3] Luo, Z., Xing, R., Huang, W., Xiong, R., Qin, L., **Ren, Y.**, Li, Y., Liu, X., Men, Y., Jiang, K., Tian, Y., Shen, G. Impacts of Household Coal Combustion on Indoor Ultrafine Particles—A Preliminary Case Study and Implications for Exposure Reduction. *International Journal of Environmental Research and Public Health*, **19**, 5161, 2022.
<https://doi.org/10.3390/ijerph19095161>
- [2] Liu, X., Shen, G., Chen, L., Qian, Z., Zhang, N., Chen, Y., Chen, Y., Cao, J., Cheng, H., Du, W., Li, B., Li, G., Li, Y., Liang, X., Liu, M., Lu, H., Luo, Z., **Ren, Y.**, Zhang, Y., Zhu, D., Tao, S. Spatially Resolved Emission Factors to Reduce Uncertainties in Air Pollutant Emission Estimates from the Residential Sector. *Environmental Science & Technology*, **55**, 4483–4493, 2021.
<https://doi.org/10.1021/acs.est.0c08568>
- [1] Luo, Q., **Ren, Y.**, Sun, Z., Li, Y., Li, B., Yang, S., Zhang, W., Hu, Y., Cheng, H. Atmospheric Mercury Pollution Caused by Fluorescent Lamp Manufacturing and the Associated Human Health Risk in a Large Industrial and Commercial City. *Environmental Pollution*, 269, 116146, 2021.
<https://doi.org/10.1016/j.envpol.2020.116146>

PRESENTATIONS

- [11] Assessing GEOS-Chem Organic Aerosol Simulations with Globally Consistent Ground-Based Measurements. The 12th International GEOS-Chem Meeting, St. Louis, MO, USA, June 2026. (Poster)
- [10] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. Atmospheric Science & Aerosols Research Webinar, University of Puerto Rico, Remote, April 2026. (Invited Talk)
- [9] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. American Geophysical Union (AGU) Annual Meeting, New Orleans, LA, USA, December 2025. (Talk)
- [8] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. Health and Air Quality Applied Sciences Team (HAQAST) Meeting, St. Louis, MO, USA, November 2025. (Poster)
- [7] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. Center for Aerosol Science and Engineering (CASE) Seminar, St. Louis, MO, USA, November 2025. (Talk)
- [6] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. International Global Atmospheric Chemistry (IGAC) Early Career Seminar, Remote, July 2025. (Invited Talk)
- [5] Progress toward Offline Aerosol Mass Spectrometry (AMS) Measurement of Organic Aerosol. Advancing Globally Distributed Air Quality Monitoring Meeting, St. Louis, MO, USA, June 2025. (Talk)
- [4] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. Advancing Globally Distributed Air Quality Monitoring Meeting, St. Louis, MO, USA, June 2025. (Talk)
- [3] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. The 42nd American Association for Aerosol Research (AAAR) Annual Conference, Albuquerque, NM, USA, October 2024. (Poster)
- [2] Emission Inventories Underestimate Black Carbon in the Global South as Revealed by Comparison of GCHP Simulations with Measurements. The 11th International GEOS-Chem Meeting, St. Louis, MO, USA, June 2024. (Poster)
- [1] Offline Aerosol Mass Spectrometry (AMS) Technique for Chemical Characterization of Filter Samples. The 4th International SPARTAN Meeting, St. Louis, MO, USA, May 2023. (Talk)

FIELD EXPERIENCE

- [4] Indoor Volatile Organic Compounds (VOCs) Sampling from Rural Residential Solid-Fuel Combustion. Jinzhong, China, November 2020–November 2022.

- [3] Second Scientific Expedition to the Qinghai-Tibet Plateau. Hoh Xil World Heritage Site, China, September–December 2019.
- [2] Atmospheric Mercury Sampling in a Fluorescent-Lamp Manufacturing Region. Zhongshan, China, July–August 2019.
- [1] Chinese National Pollution Source Census: Residential Stove Upgrade Assessment. “2+26” Cities in Northern China, July 2018–November 2019.

TEACHING EXPERIENCE

Washington University in St. Louis , Teaching Assistant	St. Louis, US
<i>EECE 514 Atmospheric Science and Climate</i>	Spring 2025
<i>EECE 202 Computational Modeling in EECE</i>	Fall 2023
<i>EECE 314 Air Quality Engineering with Lab</i>	Spring 2023
Peking University , Teaching Assistant	Beijing, China
<i>Environmental Geochemistry</i>	Fall 2019, Fall 2020

HONORS, AWARDS, AND FELLOWSHIPS

[5] Annual Research Impact Award, Washington University in St. Louis	2025
[4] Pivot 314 Fellowship, Washington University in St. Louis	2024
[3] Award for Scientific Research, Peking University	2020
[2] Outstanding Graduate, Shandong University	2019
[1] National Scholarship, Ministry of Education of China	2016, 2017, 2018

SERVICE

Reviewer	<i>Atmospheric Chemistry and Physics; npj Climate and Atmospheric Science; Environmental Science and Ecotechnology; Journal of Geophysical Research – Atmospheres</i>
Leader	Clear Spring Environmental Protection Association, Shandong University (2015–2017)